Δεύτερη Υποχρεωτική Εργασία

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Άσκηση 5

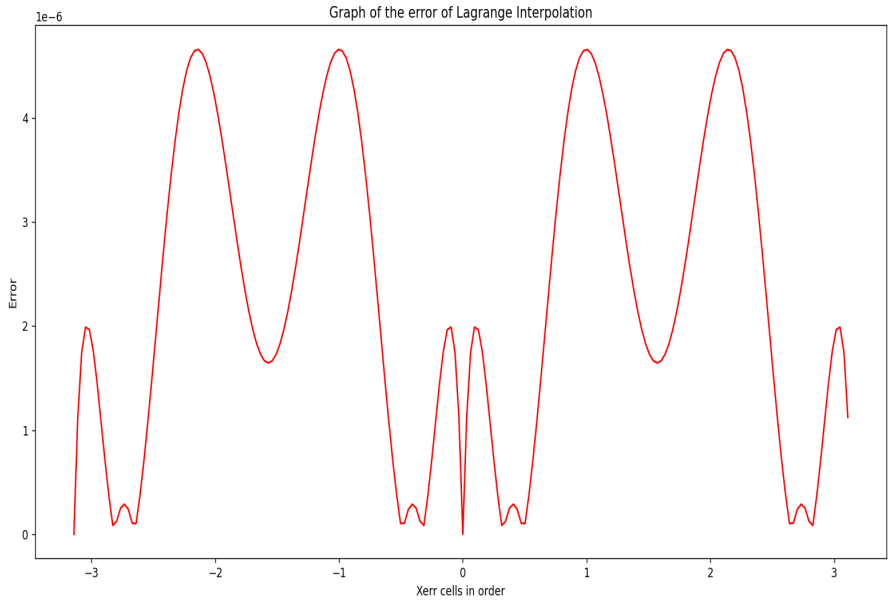
(α) Interpolating Polynomial(Lagrange form)

The Lagrange Polynomial is implemented by 2 functions. I decided create only the part of $sin(x)$ at $[0, \pi]$ (using consecutive points increasing by $\pi/9$ to split it in 9 equal subintervals up until to π that are stored in array A and sinA) in order to potentially get more precision since it's symmetric everywhere else(the function is odd with $sin(-x) = -sin(x)$ and periodic with $sin(x + 2\pi) = sin(x)$).

The function f does all the things that are needed to take care of its symmetric nature. If the input b is less than 0 it makes it positive and using a flag returns $-p(b)$ if it is. If the input b is bigger than 2π we get the remainder by the division by it, and,furthermore, if it's in the interval $[\pi, 2\pi]$ we account for the changes there too.

The function p just implements the sin function in the interval $[0, \pi]$, by utilizing the formula:
$L_i(x) = \dfrac{(x - x_0)...(x - x_{i-1})(x - x_{i+1})...(x - x_n)}{(x_i - x_0)...(x_i - x_{i-1})(x_(i - x_{i+1})...(x_i - x_n)}$
and then: $p_n(x) = \sum_{i=0}^{n} y_i \; L_i(x)$.

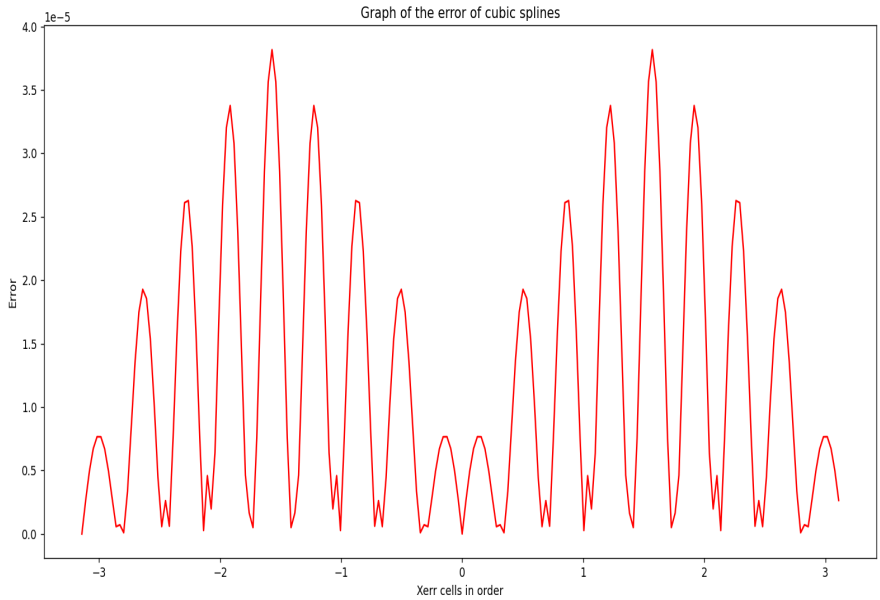
Therefore with this method we achieve 5 digits of precision, since the error is $4.658362821374773 * 10^{-6}$.



(β) Cubic Splines

I implemented the natural Cubic Splines algorithm as stated in the slides. Here too I use A and sinA from before. Here too I have the function $fspl$ handle the symmetric things(it's exactly the same as f before) and the function P create the arrays $\mathbf{A}$ and $\mathbf{r}$(but they're a little bit altered) from the theory in the slides(in general my algorithm is a little different from the slides since when I implemented it in exactly the same way as in the slides, it wouldn't produce the correct results, so I altered it a little with a bit of help from the internet). I apply $Gauss$ to it and get my c and then I calculate $a = sinA[o-1]$ with o calculated from our input(it's the upper x_i from the interval that our input is in). $b = \frac{1}{\pi/9}(sinA[o] - sinA[o-1]) - ((\frac{\pi}{9})/3)(2c[o-1,0] + c[o,0])$, $d = \frac{3}{\pi}(c[o,0] - c[o-1,0])$, $s = rad - A[o-1]$. It then returns $P(rad) = a + bs + cs^2 + ds^3$.

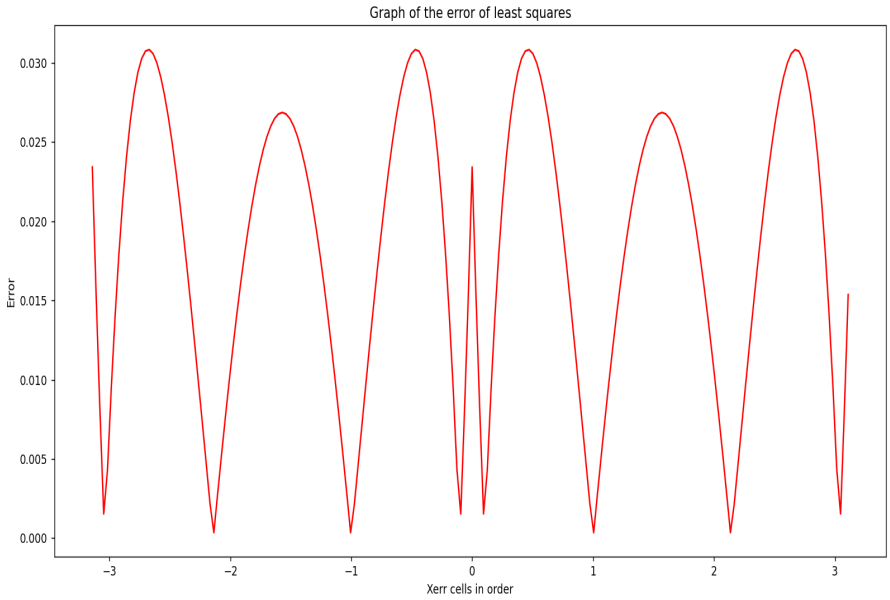
Therefore with this method we achieve 4 digits of precision, since the error is $3.8160377358487096 * 10^{-5}$.



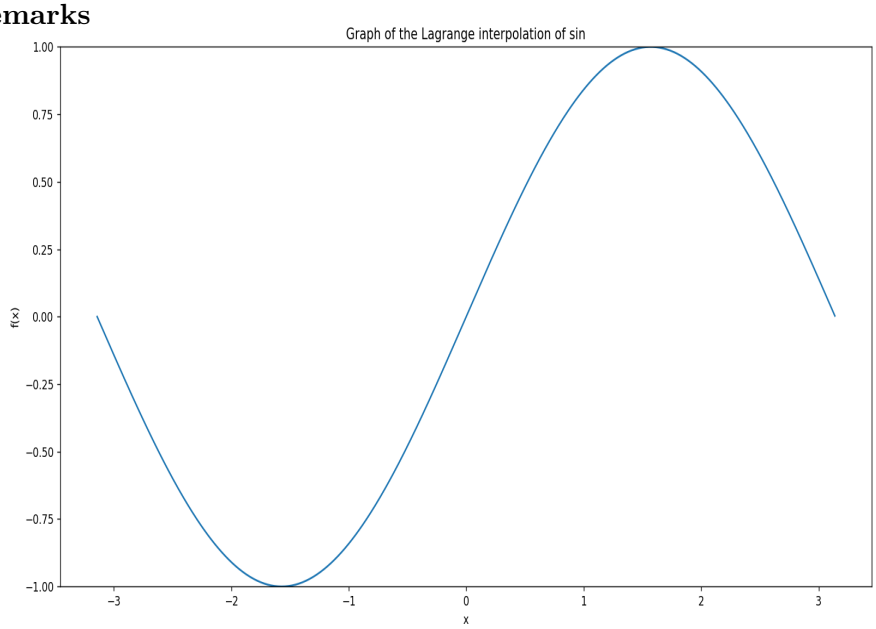
(γ) Least Squares method(with 3rd degree polynomial)

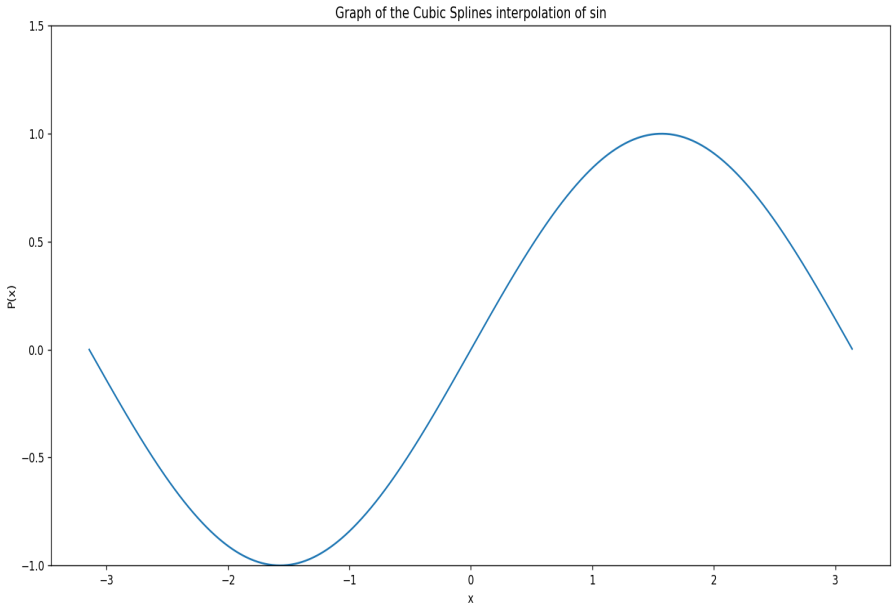
I implemented the Least Squares method exactly as stated in the slides(except for the fact that I use Gaussian Elimination instead of the inverse of $X^T X$). Here too I have the function fls handle the symmetric things(it's exactly the same as f before) and the function $least_squares$ create the Vandermonde matrix $\mathbf{X}$ and the matrix $\mathbf{y}$ from the theory in the slides. It then finds X^T using the $transposer$ function that I created. It then multiplies $X = X^T X$ and $y = X^T y$ and then it uses $\beta = Gauss(X, y)$. Lastly It returns $\beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$ (x being the input of the function).

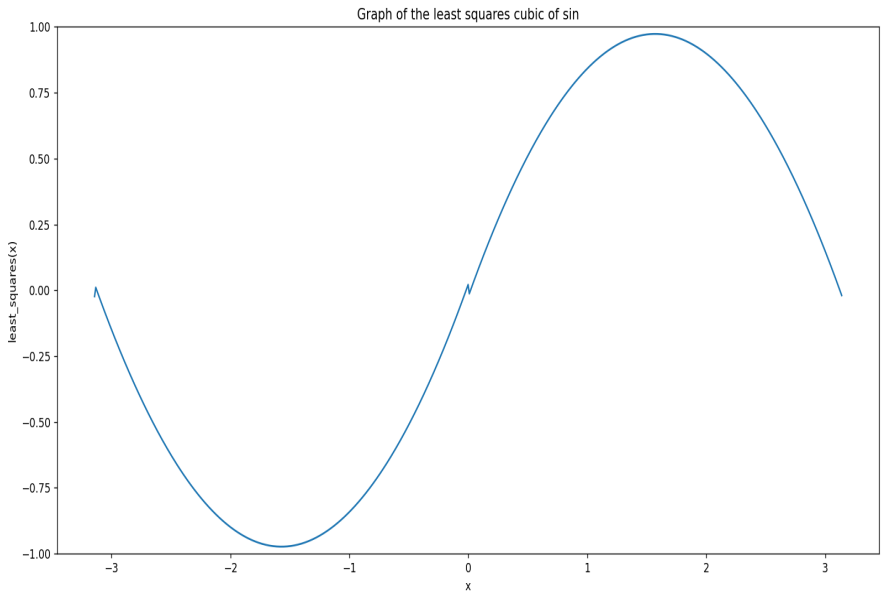
Therefore with this method we achieve ≈2 digits of precision, since the error is $3.083121389682053 * 10^{-2}$.


Graph of the error of least squares
Error
Xerr cells in order
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−3
−2
−1
0
1
2
3

* Remarks


Graph of the Lagrange interpolation of sin
f(x)
x
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−3
−2
−1
0
1
2
3





For each of the functions, I plot them over the 200 points in $[-\pi, \pi]$, because that's the best way to see if they're correct or not. I also print out their error over these 200 points in a plot using the functions inf_normf, inf_spl and inf_normls. These functions also find the maximum error in $[-\pi, \pi]$ for each function and print it. Note that for the Least Squares method we get the least precision and this shows also from the visible bumps in at $-\pi$ and 0, namely the places where fls uses the symmetric properties of the function $sinx$ to connect the points.

Άσκηση 6

(α) Trapezoid Rule

I first create A and sinA with $\frac{\pi}{2}$ as the interval so for 11 equally spaced points we get $\frac{\pi}{2}/10 = \frac{\pi}{20}$ as the size of our intervals.

I implement this method exactly as instructed, with the function *trapezoid_rule* first computing $S = \sum_{i=1}^{N-1} f(x_k)$ and then returning $\frac{\beta - \alpha}{2N}(f(x_0) + f(x_N) + S)$

The theoretical error is $e = \frac{\frac{\pi}{2}^3}{12N} M = 3.2298204875312306 * 10^{-3}$

The arithmetical error is $20570136456428134 * 10^{-4}$

(β) Simpson Rule

I implement this method exactly as instructed, with the function *Simpson_rule* first computing $S1 = \sum_{i=1}^{N/2-1} f(x_{2i})$ then, $S2 = \sum_{i=1}^{N/2} f(x_{2i-1})$ and then returning $\frac{\beta - \alpha}{3N}(f(x_0) + f(x_N) + 2S1 + 4S2)$

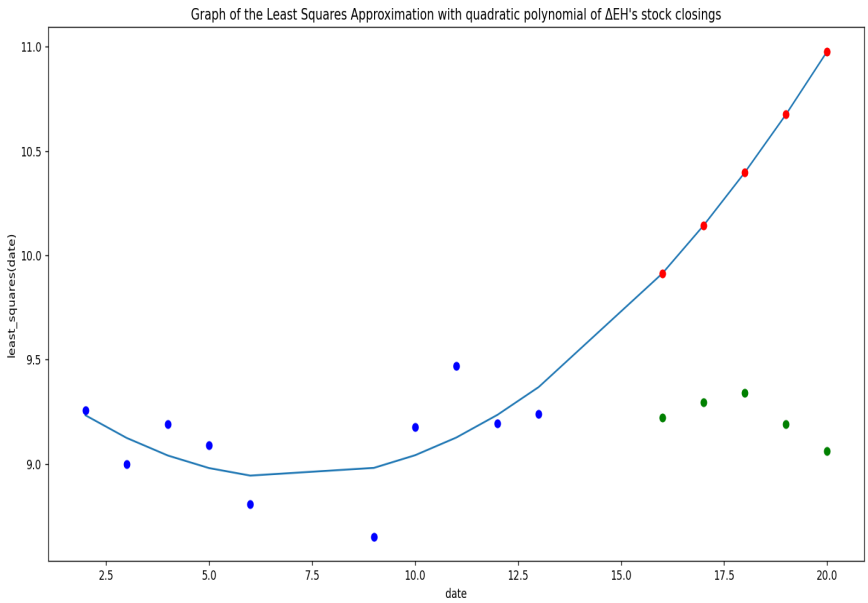
The theoretical error is $e = \frac{\frac{\pi}{2}^5}{180N^4} M = 9.10981095635197 * 10^{-13}$

The arithmetical error is $3.3922209004000337e - 06 * 10^{-6}$

Άσκηση 7

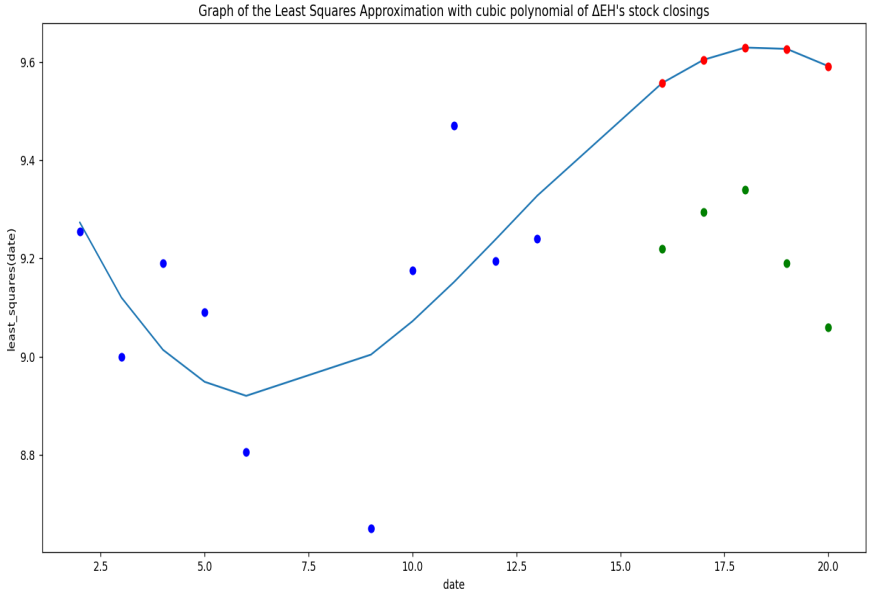
1. Value of daily stock closing of ΔΕΗ:
   My birthday is at 15/10 so the days that I choose are [2, 3, 4, 5, 6, 9, 10, 11, 12, 13]. In each case, after I make the model according to these data points, I then predict the closings for 5 days after, namely [16, 17, 18, 19, 20]. Also the only difference between the quadratic, cubic and quartic implementations is the number of terms in the polynomials. All the data were taken from: http://www.capital.gr/finance/el/allstocks for the stocks of OTE and ΔΕΗ.

   (α) QUADRATIC:



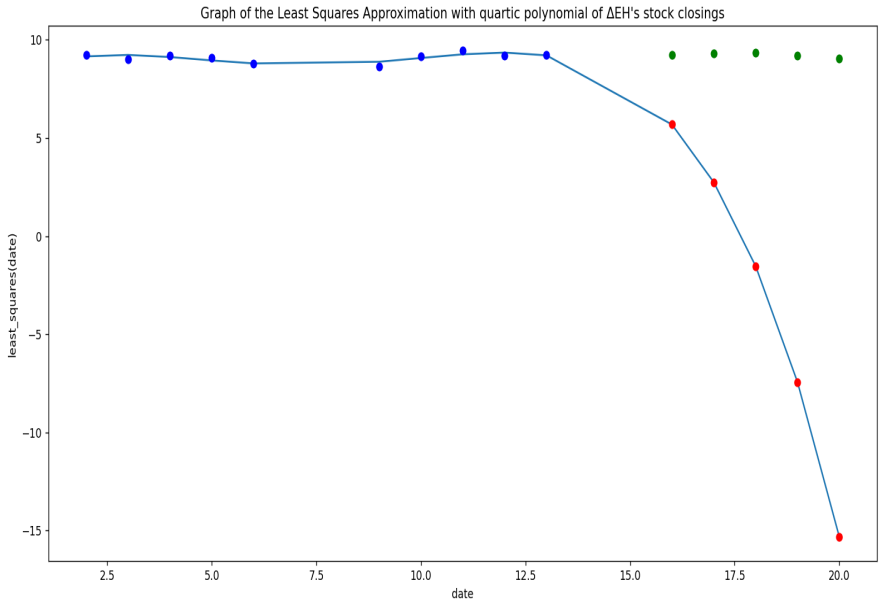
The points that we already know are plotted with blue color.
The points that we want to predict are plotted with red color.
The real values of the stock closes after(and including) the 16th are plotted in green.
Prediction for 16 /10/2023: 9.9131
Real value: 9.22
Prediction for 17 /10/2023: 10.1431
Real value: 9.295
Prediction for 18 /10/2023: 10.3973
Real value: 9.34
Prediction for 19 /10/2023: 10.6756
Real value: 9.19
Prediction for 20 /10/2023: 10.9782
Real value: 9.06

(β) **CUBIC:**



The points that we already know are plotted with blue color.
The points that we want to predict are plotted with red color.
The real values of the stock closes after(and including) the 16th are plotted in green.
Prediction for 16 /10/2023: 9.5573
Real value: 9.22
Prediction for 17 /10/2023: 9.6048
Real value: 9.295
Prediction for 18 /10/2023: 9.6297
Real value: 9.34
Prediction for 19 /10/2023: 9.6271
Real value: 9.19
Prediction for 20 /10/2023: 9.5918
Real value: 9.06

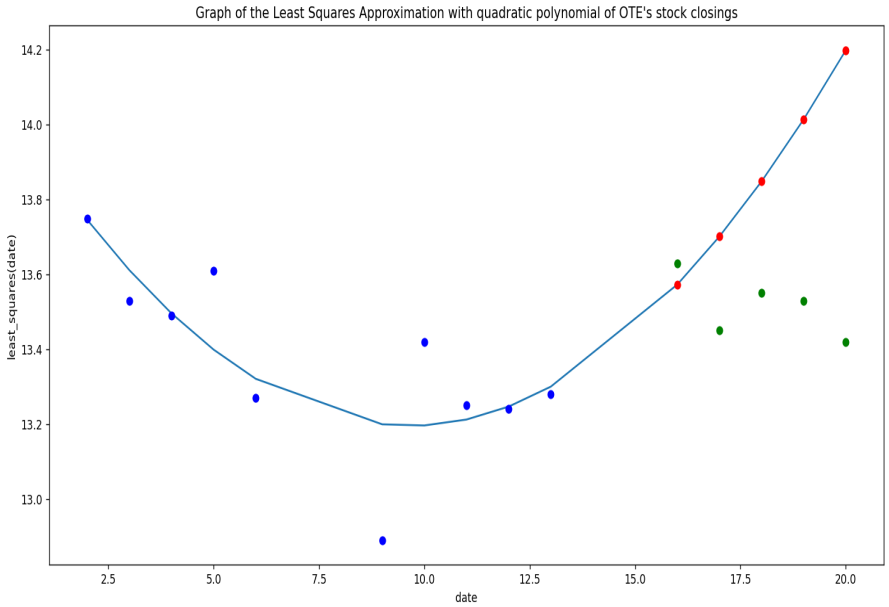
(γ) QUARTIC:



The points that we already know are plotted with blue color.
The points that we want to predict are plotted with red color.
The real values of the stock closes after(and including) the 16th are plotted in green.
Prediction for 16 /10/2023: 5.6886
Real value: 9.22
Prediction for 17 /10/2023: 2.7463
Real value: 9.295
Prediction for 18 /10/2023: -1.5225
Real value: 9.34
Prediction for 19 /10/2023: -7.4298
Real value: 9.19
Prediction for 20 /10/2023: -15.318
Real value: 9.06

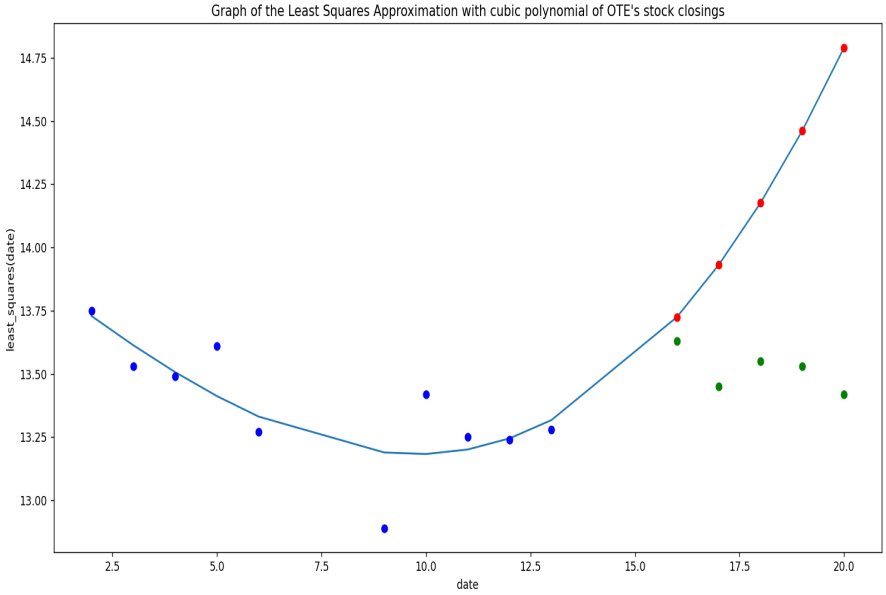
2. Value of daily stock closing of OTE:

   (α) QUADRATIC:



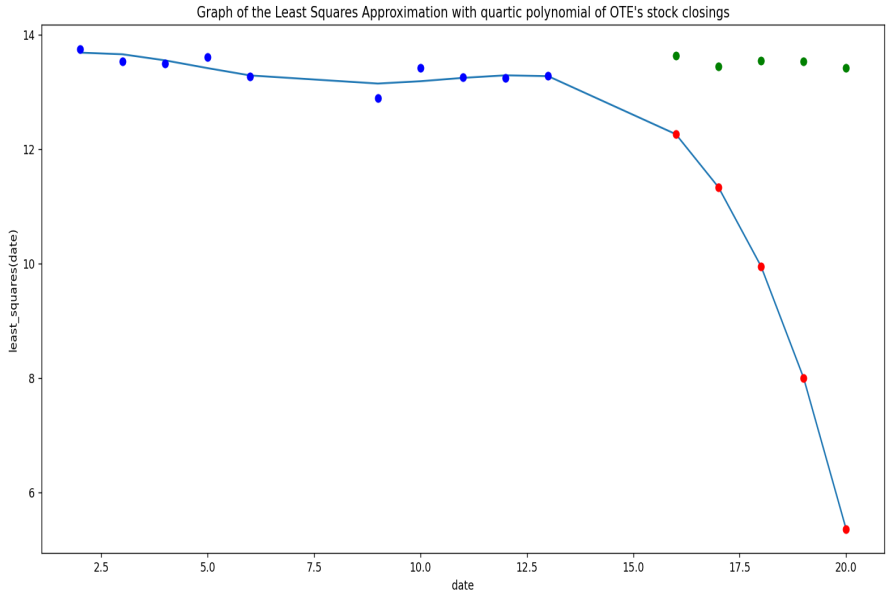
The points that we already know are plotted with blue color.
The points that we want to predict are plotted with red color.
The real values of the stock closes after(and including) the 16th are plotted in green.
Prediction for 16 /10/2023: 13.5729
Real value: 13.63
Prediction for 17 /10/2023: 13.7013
Real value: 13.45
Prediction for 18 /10/2023: 13.8484
Real value: 13.55
Prediction for 19 /10/2023: 14.0144
Real value: 13.53
Prediction for 20 /10/2023: 14.1991
Real value: 13.42

(β) **CUBIC:**



The points that we already know are plotted with blue color.
The points that we want to predict are plotted with red color.
The real values of the stock closes after(and including) the 16th are plotted in green.
Prediction for 16 /10/2023: 13.7245
Real value: 13.63
Prediction for 17 /10/2023: 13.9307
Real value: 13.45
Prediction for 18 /10/2023: 14.1756
Real value: 13.55
Prediction for 19 /10/2023: 14.4613
Real value: 13.53
Prediction for 20 /10/2023: 14.79
Real value: 13.42

(γ) QUARTIC:



The points that we already know are plotted with blue color.
The points that we want to predict are plotted with red color.
The real values of the stock closes after(and including) the 16th are plotted in green.
Prediction for 16 /10/2023: 12.2592
Real value: 13.63
Prediction for 17 /10/2023: 11.333
Real value: 13.45
Prediction for 18 /10/2023: 9.9516
Real value: 13.55
Prediction for 19 /10/2023: 8.001
Real value: 13.53
Prediction for 20 /10/2023: 5.3553
Real value: 13.42

We can see that the one with the best predictions is the cubic and quadratic (it's situational for which one is better), followed by the quartic.

Παρακαλώ να δείτε και τα σχόλια στον κώδικα που έχω γράψει, καθώς μπορεί να περιέχουν ορισμένες περεταίρω εξηγήσεις.